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Executive functioning and relational framing in verbal adults☆

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ABSTRACT

Relational Frame Theory (RFT) sees the operant acquisition of various patterns of derived relational responding as being key to language and cognition. Executive function (EF) is a concept widely used in cognitive psychology and is said to be critical for problem-solving, self-regulation, and overall success and wellbeing. The current study was a preliminary attempt to study the relationship between a measure of EF and fluency in derived relational responding. Forty-two verbal adults (25 males) participated in the study. All of them undertook a standardized EF test, the Wisconsin Card Sort Test (WCST) and an experimenter-designed arbitrary relational responding assessment. The WCST required participants to match compound stimuli based on unspecified, frequently changing rules. The relational assessment evaluated rate of correct derived relational responding across seven relational frames. The results showed strong negative correlations between perseverative errors on the WCST and the overall rate of correct responses on the relational assessment, but no such correlations between correct responses on the WCST and those on the relational assessment.

1. Executive functioning and relational framing in verbal adults

Executive function (EF) is a concept widely used in cognitive psychology and refers to one type of self-regulatory process involved in problem-solving (Hofmann et al., 2012; Roebers, 2017). It is said to include a variety of different capacities that enable purposeful, goal-directed behavior, including working memory (temporarily store, process, and manipulate information), inhibitory control (inhibit inappropriate impulses and control one's behaviors and thoughts), cognitive flexibility and set-shifting (change strategies and rapidly switch from one task to another), planning (develop an effective approach to achieving goals), and attentional control (selectively attend to relevant stimuli). Measures of EF have been shown to correspond with academic achievement and emotion regulation and predict well-being (Liberman, 2007; Morgan et al., 2018; McClelland et al., 2015; Toh et al., 2020). Individuals with schizophrenia, attention deficit hyperactivity disorder, and psychosocial challenges such as anxiety tend to have difficulties during EF tasks (Brown, 2007; Orellana & Slachevsky, 2013; Zhou et al., 2022). EF is widely regarded to be a critical area of research and practice within cognitive psychology.

Cognitive psychologists posit that EF comprises of several processes including set-shifting. Set shifting, also described as cognitive flexibility, refers to quickly and efficiently switching between mental sets or tasks in order to adapt to a changing environment (Irwin et al., 2019; Pa et al., 2010). For example, when a road sign indicates a change in speed limit, we adapt our behavior to drive within that speed, and we repeatedly adjust our driving behavior in accordance with changing road signs. Upon encountering a new sign, driving at the previous speed limit is no longer acceptable and may result in a speeding ticket. In this example, individuals adapt to a changing set of rules specified clearly by the road signs.

In other instances, individuals may have to adapt to new rules that are not specified clearly but need to be discerned from changing schedules of reinforcement and punishment. Consider for instance, if one is in a foreign country asking a stranger for directions to the nearest ATM to withdraw cash. If saying "ATM" gets no response, we might try a different approach such as saying "money" or using a translation application on our phone, or finding an English-speaking individual, or any of several other options. Although saying "ATM" may have resulted in reinforcement in the past in an English-speaking country, we adapt our behavior in accordance with the changing schedules of reinforcement in the foreign country. Simply saying "ATM" repeatedly is

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rendered useless in the new country. Set-shifting is said to be critical for problem-solving and creative "out-of-the-box" thinking (Cooper & Marsh, 2016; Cronin-Golomb et al., 1994). Critically, set-shifting may be seen as a potential bridge between the domains of cognitive psychology and behavior analysis. Although EF and its components are defined in mentalistic terms, there are specific behavioral effects of set-shifting which behavior analysts have been studying for decades in terms of rule-governed behavior and contingency insensitivity (O'Connor et al., 2018; Ruiz et al., 2020; Grant & Cassidy, 2022).

There are several validated measures of set-shifting that are available for children and adults (see Diamond, 2014, for an overview of measures available). In the current study, we used the Wisconsin Card Sort Test (WCST), a standardized test that has been normed for persons aged between 6½ through 89 years of age. Several studies have shown support for the construct validity of this test as a measure of EF (e.g., Heaton, 1993). During the WCST, individuals are required to match a sample compound stimulus to one of four comparison compound stimuli based on a sorting rule (e.g., color). The rules are not explicitly stated; but are implied through either negative or positive feedback for each matching response. Furthermore, after several trials, the sorting rule is changed without the participant's knowledge. Numbers of correct matching responses, errors, and perseverative errors (i.e., matching in accordance with a pre-existing rule even after receiving negative feedback) are recorded during the test. Similar work has been conducted in behavior analysis using the Implicit Relational Assessment Procedure (IRAP; see O'Toole, Barnes-Holmes, Murphy, O'Connor, & Barnes-Holmes, 2009; Finn et al., 2017) The IRAP requires participants to respond accurately and rapidly to a series of relational tasks. In some blocks of tasks, participants are asked to respond in a way that is consistent with previously learned relations (Is Spring before Summer = True); and in other blocks, responding in an inconsistent pattern is required (Is Spring before Summer = False). Key differences between prior work using the IRAP and the WCST are that participants are not informed about how to respond in the latter, and there are more than two ways in which participants can respond (i.e., they pick one of four options during each trial on the WCST as opposed to two options on the IRAP).

While EF and set-shifting have been studied extensively by cognitive and clinical psychologists, these are not technical terms within behavior analysis and the topic has received relatively little attention (Kelly, 2020). Hayes et al. (1996) have provided a theoretical behavior analytic account describing EF (and other psychological constructs) as a subset of rule-governed behavior. They noted that from a relational frame theory perspective, EF tests such as the WCST aim to investigate the conditions under which people select among available rules or generate new ones when they no longer work.

Relational Frame Theory (RFT), a behavior analytic theory of language and cognition defines rule-governed behavior as verbal antecedent stimuli specifying the causal relations between stimuli and events (Hayes et al., 2001). The verbal rules that are generated are said to be a product of derived relational responding, whereby verbally constructed consequences and experiences with contingencies transform the function of related events. Individuals relate events in a variety of ways such as frames of coordination (e.g., the spoken word "cat" is the same as an actual cat), distinction (e.g., cats are "different" from dogs), comparison (e.g., something being "more" valuable than something else), hierarchy (e.g., something or someone belonging to a category, group, or class), temporality (e.g., Fran arrived "after" Tina to the party), spatiality (e.g., an object being "over" the other), and deictics (e.g., relating from the perspective of "I").

Although neurotypical individuals readily demonstrate derived relational responding (e.g., Colbert et al., 2017), there have been some reports of neurodivergent adults needing additional support to facilitate this repertoire (e.g., Zagrabska-Swiatkowska et al., 2020). Similarly, some neurodivergent individuals have been shown to have difficulties with set-shifting (Bacelo & Knight, 2002; Boshomane et al., 2021; Faustino et al., 2021; Miller, Ragozzino, Cook, Sweeney, & Mosconi, 2014; Yasuda et al., 2014). On balance, others have reported no significant difference in perseverative errors on the WCST in autistic children after controlling for verbal functioning (Liss et al., 2001), suggesting that perseverative tendencies observed during the WCST may be related to verbal ability. If this were the case, we may expect to see correlations between scores on the WCST and verbal ability. Indeed, RFT views derived relational responding and processes underlying language as being synonymous (see Dymond & Roche, 2013 for a book-length review). Thus, from an RFT perspective, set-shifting may be seen to involve derived relational responding, and individuals having challenges with set-shifting (i.e., those demonstrating an insensitivity to contingencies) may also demonstrate lower levels of correct derived relational responding and psychological flexibility.[1]

A handful of previous studies have assessed set-shifting using measures such as the WCST and its relationship with self-reported measures of psychological flexibility in adults (Grant & Cassidy, 2022; O'Connor et al., 2018). These studies found positive correlations between self-reported measures of psychological flexibility and performance-based measures of rule-following on some EF tasks. Specifically, they showed that sensitivity to direct contingencies may be diminished when some rules become over-generalized. Tyrberg and colleagues (2021) recorded the verbal behavior emitted by eleven adult participants during the WCST and coded these statements for patterns of relational frames. Specifically, these authors identified relational cues (e.g., "after", "more", "I") in participants' verbal behavior, and found that deictic and temporal framing dominated participants' verbal behavior during critical time points including when sorting rules changed during the WCST.

One could argue that during rule changes, relational framing is particularly relevant because responding according to a previous rule results in negative feedback, and one must identify the new rule in effect. Critically, however, none of the aforementioned studies assessed participants' overall derived relational responding independent of the WCST. If set-shifting indeed involves relational framing, an important first step for a research program in this domain is to identify associations, if any, between scores on the WCST and derived relational responding measured using a relational assessment. Investigating set-shifting from a functional-cognitive framework bridges the behavioral and cognitive science divide (de Houwer, 2011; de Houwer et al., 2017; Barnes-Holmes & Hussey, 2016). Such work is already being carried out extensively by researchers studying the impact of training derived relational responding on intelligence thereby offering a functional analytic theory to examine the concept of intelligence (Cassidy et al., 2015). In terms of EF, and specifically set-shifting, behavior analysts can make important contributions including identifying the environmental variables that cause and moderate set-shifting, as well as offer training protocols to increase efficient set-shifting. In doing so, we will expand our understanding of EF and the learning histories that facilitate set-shifting, gain insight into how behavior analysts can support the development of this repertoire, and afford better explanations and theorizing to both the cognitive approach and the functional approach. Considering RFT as a theory of language *and* cognition, such work has the potential to advance the scope and depth of RFT.

In the current study, we chose to measure participants' fluency in derived relational responses across several response topographies as a metric for the relational assessment. Specifically, we attempted to measure responses that involved stating a relational cue (e.g., "more") when asked to relate two stimuli vs. stating the relata (e.g., saying "red"

[1] Psychological flexibility is defined as the nonjudgemental acceptance of all psychological events, even if uncomfortable. Interventions based on RFT to promote psychological flexibility teach individuals to frame uncomfortable events as 'only a part of me' and under the control of changing contingencies instead of unchanging literal truths (Luciano et al., 2012; Grant & Cassidy, 2022).

when asked "which one is more?") vs. responding with "yes/no" to a relational statement (e.g., saying "yes" when asked, "Is green more than red?"). Identifying if there are differences in fluency in derived relational responding across response topographies and whether these differences matter in verbal adults could be an important question given the differences noted in children (see Horne et al., 2004; Greer, 2020; Sivaraman et al., 2023, 2025, for differences in relational responding across topographies in children). Specifically, we may expect yes/no responses to relational statements to be more challenging than other topographies based on prior research (e.g., Maloney & Barnes-Holmes, 2016; Maloney et al., 2020). Yes/no responding may be seen as a "higher" level of relational responding because the trueness or falseness of a relational statement can only be determined after the relation has been acquired to some basic level (Barnes-Holmes et al., 2020; see also Sivaraman et al., 2025 for differences between yes/no responses and other topographies in relational responding in children). That is, responding with yes/no to the statement, "A nickel is more valuable than a dime" is not feasible until the cue "more valuable" comes to control the relation between the two coins. The current study attempted to evaluate if there were such differences in rate of correct derived relational responding across various speaker response topographies. Furthermore, we used the Relational Evaluation Procedure (REP; see Barnes-Holmes et al., 2001; Corbett et al., 2017; Stewart et al., 2004, for more on the REP) for the assessment. This methodology allows participants to respond to stimulus relations defined by various sets of contextual cues on a given trial.

In the current study, we administered the WCST and a relational assessment, similar to the one in Kirsten and Stewart (2021) to 42 verbal adults. This was a preliminary investigation into the following research questions: (a) whether there was an association between performance on the WCST (e.g., total trials, total correct responses, and perseverative errors) and the overall rate of correct derived relational responding during the assessment, (b) if there was a correlation between scores on the WCST with rate of correct derived relational responding on specific relational frames (i.e., coordination, distinction, comparison, temporality, hierarchy, spatial, and deictic) during the relational assessment, and (c) whether there were any significant differences between the topographies of responses on the relational assessment.

2. Method

2.1. Participants

Participants included 42 consenting verbal adults (27 males, 15 females; 33 self-identifying as neurotypical, 9 self-identifying as neurodivergent) between the ages of 18–40 years ($M = 28$). All participants received a $20 Amazon gift card for successfully completing the study. We recruited all our participants by circulating a flyer about the study on social media (i.e., Facebook and Instagram). Table 1 lists the complete demographic summary of all participants. The neurodivergent participants self-reported as having autistic characteristics ($N = 3$), characteristics of ADHD ($N = 3$), a learning disability ($N = 1$), or social anxiety ($N = 2$). We. We did not verify their diagnostic reports or conduct any diagnostic tests as part of the study. An *apriori* power analysis using G-power indicated that the necessary sample size to achieve 80 % power to detect a medium effect ($\rho = .4$) at an alpha level of .05 was 44.

Table 1
Participant demographic information.

		Total sample
Number	42	
Gender	Male	27
	Female	15
Mean Age (SD)		28.87 (5.64)
Ethnicity		18 Asian, 21 Black, 2 White, 1 Hispanic
Self-reported neurodivergence status	Neurotypical	33
	Neurodivergent	9
Mean years of education		16.6

2.2. Setting and materials

The materials used in this study included a computer, an iPad, a timer, the Wisconsin Card Sorting Test (WCST), and the arbitrary relational assessment developed by the experimenters. The first author administered the online-version of the WCST and Microsoft PowerPoint for the arbitrary relational assessment. In addition, the researcher used an iPad to record data and Zoom to conduct sessions.

2.3. Measures

2.3.1. Wisconsin Card Sorting Test

The WCST required the participants to match a sample compound stimulus to one of four comparison compound stimuli based on a sorting rule (e.g., color, Berg, 1948). For each trial, there were four comparison compound stimuli on the top row of the computer screen and a sample compound stimulus on the bottom row of the screen. Correct and incorrect responses were automatically recorded by the computer program. During the WCST, we collected the following data: the total number of trials, the total number of correct responses, the number of perseverative errors (calculated by summing the number of responses that involved perseveration of a previously successful strategy despite negative feedback signaling that the rules of the test had changed), and the number of perseverative responses as defined in the WCST manual. Perseverative responses are persistent responses made by a participant on the basis of an incorrect (previous or novel) stimulus dimension (Heaton, 1993). For example, a participant may attempt to sort a card based on its form, receive feedback that this response is incorrect, and yet continue to make the same mistake on the following card. Typically, perseverative responses are incorrect (i.e. the response does not match the correct sorting rule), and include perseverative errors (Berg, 1948). However, a perseverative response may also be correct because of the potential for ambiguous responses. On some trials of the WCST, the chosen stimulus card may match the response card on multiple dimensions (e.g. the response card 'four green circles' matches the stimulus card 'four blue circles' on both number and form) making it unclear which sorting rule was being applied by the participant. Responses on such trials were termed ambiguous responses. Perseverative responses include perseverative errors plus ambiguous responses. Miles et al. (2021) recently recommended the use of both perseverative errors and perseverative responses as a measure of set-shifting when using the WCST.

2.3.2. Arbitrarily applicable derived relational responding assessment

The relational assessment evaluated seven specific relational frames: coordination, distinction, comparison, temporality, hierarchy, spatiality, and deictic relations across six tests (adapted from Kirsten & Stewart, 2021). An adaptation of the REP format was used in the assessment. The relational stimuli used were simple, monochromic shapes (circles, triangles, squares) separated by a single Latin letter indicating a contextual cue (i.e., *S* for Same, *D* for Different, *M* for More, *L* for Less, *B* for Before, *A* for After, *C* for Contains, and *I* for Inside, *O* for Over and *U* for Under). The only differences in our assessment to that of Kirsten & Stewart, 2021, was that we included three distinct trial types, i.e., speaker-relation responses, speaker-relata responses, and speaker yes/no responses for each frame, and we assessed for deictic framing. Prior research has typically only included either listener or speaker responses, but we included these distinct trial types to better capture the extent of relational responses emitted by individuals in everyday settings (see Sivaraman et al., 2025 for a discussion on response topographies).

All relations were tested independently except for coordination and distinction which were combined in the assessment. With the exception of the deictic frame, we assessed all other frames using 30 trials, i.e., 10 trials each involving speaker-relation responses, speaker-relata responses, and speaker yes/no responses. Each trial involved visual stimuli representing a relational network and a verbal question read out by the researcher. We adapted the deictics assessment based on Rehfeldt et al. (2007), and the assessment consisted of 48 trials and evaluated the participant's performance on simple relations (12 trials), reversed relations (24 trials), and double-reversed relations (12 trials) involving the vocal cues I-You, Here-There, and Now-Then relations. For the deictic frame, all trials involved a verbal statement and question read out by the researcher. We had fewer trials than Rehfeldt et al. (2007) in the interest of time.

We added the number of correct responses for each frame divided by the duration to conduct all trials in that frame to report rate of correct responses. Duration was onset when the researcher began the first trial for a frame and offset when the participant responded to the last trial in that frame. We also reported the overall rate of correct responding for the entire relational assessment.

2.4. Procedure

2.4.1. General procedure

We administered both the WCST and the relational assessment within one Zoom session, which lasted between 40 and 65 min.

2.4.2. Wisconsin Card Sorting Test

Before beginning the WCST test, the researcher provided the following scripted instruction, "your task in this assessment is to match the bottom card with one of the cards on top [showing a sample trial arrangement]. There is a specific rule you need to follow for matching, but I am not allowed to tell you the rule. The computer will give you feedback on your responses. The rules can change without notice several times during the session". Once the participant acknowledged that they understood the instruction, they were granted control of the computer, using Zoom's "remote control" feature, allowing them to navigate the test and emit matching responses independently. The test automatically switched to a new rule after the participants emitted 10 consecutive correct matching responses. That is, the participants did not receive any information or signal that the sorting rule had changed. The test involved 3 rules, match by color, shape, or number of stimuli, and each rule repeated once in that same order, for a maximum of six rules and 128 trials. The system automatically stopped trial-presentation after a participant completed all six rules with 10 consecutive correct responses or reached the maximum number of trials.

2.4.3. Arbitrarily applicable derived relational responding assessment

For each frame, the researcher began this assessment (henceforth referred to as the *relational assessment*) with a scripted instruction. For example, in the coordination and distinction test, the researcher explained, "We are going to talk about things that are the same and different. I will show you two colored shapes with a letter between them. The letter represents the relation between the colors. In this session, 's' stands for 'same' and 'd' stands for 'different'." The same instruction was adapted for each frame. For each trial, the researcher presented the visual stimulus and its scripted verbal statement. For example, the researcher presented [green circle] [S] [orange circle] along with the verbal statement of the network, "green is the same as orange". Questions became increasingly difficult and required directly trained, mutually entailed, and combinatorially entailed relational responses.

In the speaker-relation trials, the participant verbally identified the relation between two given stimuli. In the speaker-relata trials, the participant verbally identified the stimulus when a relational cue was specified. During a speaker-yes-no trial, the participant had to respond with either "yes" or "no" to a verbal statement. For example, given the relational network "blue is more than green, blue is less than yellow", for a speaker-relation trial, the researcher asked, "what is green to yellow – more or less?" During a speaker-relata trial, the researcher asked, "which one is less – green or yellow". For a yes-no trial, the researcher asked, "Green is less than yellow, yes or no?". For each frame, we conducted six directly-trained and mutually-entailed trials each (two of each trial-type), and eighteen combinatorially-entailed trials (six of each trial-type). See Table 2 for a full list of trial arrangements for each frame.

The deictics assessment was conducted in the exact same manner as Rehfeldt et al. (2007) albeit with fewer trials. We included 12 directly trained trials, 24 single reversed trials, and 12 double reversed trials. Prior to the deictics frame, the researcher explained to the participant that they would hear a scenario and then be asked a question based on the scenario. For instance, the researcher might say, "I have a water bottle here in New York, and you have a phone there in Florida." A single reversed question involved "If I were you and you were me, what would you have?". A double reversed question involved "If I were you and you were me and I, here was there and there was here. What would I have?".

During the relational assessment, the researcher offered praise for every other response (e.g., "great job"), regardless of accuracy. For each trial, the researcher allowed approximately 5 s for the participant to respond. If there was no response, the researcher repeated the trial. If there was still no response or the participant indicated they didn't know the answer or they emitted an incorrect response, the researcher moved on to the next trial.

2.5. Interrater reliability and procedural fidelity

The second author served as the second observer and recorded data and procedural fidelity for fifteen participants' assessments (approximately 36 % of sessions). Both observers were trained behavior analysts with ten years of experience collecting trial-by-trial behavioral data in children and adults. Both observers had access to a sheet with all the correct responses to the relational assessment. The correct responses were determined and agreed upon by all three authors. Both observers used that answer sheet to score each response as being correct or incorrect. We calculated trial-by-trial interobserver agreement, i.e., for each trial we recorded whether there was agreement between the two observers (e.g., both observers scored the trial to be correct or incorrect), and we divided the total trials with agreement by the total number of trials. We found reliability to be 98.1 % (range, 95.5–100 %) for the relational assessment indicating an overall high reliability (Reed & Azulay, 2011). All data collected during the WCST was automated through the digital administration system. Procedural fidelity was found to be 100 %. To calculate procedural fidelity, for each trial, we observed whether (or not) each trial in the relational assessment was administered according to the scripted instruction sheet which contained the antecedent instruction to be spoken by the researcher.

2.6. Split half reliability of the relational assessment

We calculated split-half reliability to evaluate the internal consistency of each test in the relational assessment. We split each test (each relational frame) into two halves, i.e., into odd-numbered trials and even-numbered trials. We found the Spearman-Brown corrected coefficient to be .87 for coordination/distinction, .92 for comparison, .92 for temporality, .83 for hierarchy, .84 for spatial, and .95 for deictic trials.

2.7. Data analyses

Considering the low sample size, we conducted Spearman's nonparametric correlational analyses to answer our first two research questions (i.e., associations between measures obtained from the WCST and the relational assessment). For all correlation coefficients, we used the following descriptors: weak (range: .1–.4), moderate (range: .4–.6), and strong (range: .6–.9). For our final research question involving

Table 2
Trial descriptions for the relational assessment.

Frame	Sample network	Directly trained	ME	CE
Coordination (cue: same)	Green is the same as orange; orange is the same as pink	Speaker-relata: What is green the same as? Speaker-relation: Green and orange - are they the same or different? Speaker-yes-no: Green is the same as orange, yes or no?	Speaker-relata: What is orange the same as? Speaker-relation: Orange and green - are they the same or different? Speaker-yes-no: Orange is the same as green, yes or no?	Speaker-relata: What is pink the same as? Green or nothing? Speaker-relation: Green and pink - are they the same or different? Speaker-yes-no: Pink is the same as green, yes or no?
Distinction (cue: different)	Yellow is different from red, red is different from blue	Speaker-relata: What is yellow different from? Speaker-relation: Yellow and red- are they the same or different? Speaker-yes-no: Yellow is different from red, yes or no?	Speaker-relata: What is red different from? Speaker-relation: Red and yellow- are they the same or different? Speaker-yes-no: Red is different from yellow, yes or no?	Speaker-relata: What is blue different from? Yellow or nothing? Speaker-relation: Yellow and blue- are they the same or different? Speaker-yes-no: Blue is the same as yellow, yes, no or don't know?
Comparison (cues: more/less)	Blue is more than green, green is more than yellow	Speaker-relata: What is more? Blue or green? Speaker-relation: Is blue more or less than green? Speaker-yes-no: Blue is more than green, yes or no?	Speaker-relata: What is more? green or blue? Speaker-relation: Is green more or less than blue? Speaker-yes-no: Green is more than blue, yes or no?	Speaker-relata: What is less? Blue or yellow? Speaker-relation: Is blue more or less than yellow? Speaker-yes-no: Blue is more than yellow, yes or no?
Temporality (cues: before/after)	Yellow is before red, red is before blue	Speaker-relata: What is before? Yellow or red? Speaker-relation: Is yellow before or after red? Speaker-yes-no: Yellow is before red, yes or no?	Speaker-relata: What is before? Red or yellow? Speaker-relation: Is red before or after yellow? Speaker-yes-no: Red is before yellow, yes or no?	Speaker-relata: What is before yellow or blue? Speaker-relation: Is yellow before or after blue? Speaker-yes-no: Yellow is after blue, yes or no?
Containment (cues: inside/contain)	Yellow is inside red; red is inside blue	Speaker-relata: Which one is inside? yellow or red? Speaker-relation: Is yellow inside, or does it contain red? Speaker-yes-no: Yellow is inside red, yes or no?	Speaker-relata: Which one is inside? Red or yellow? Speaker-relation: Is red inside or does it contain yellow? Speaker-yes-no: Red is inside yellow, yes or no?	Speaker-relata: Which one is inside? Yellow or blue? Speaker-relation: Is blue inside or does it contain yellow? Speaker-yes-no: Yellow is inside blue, yes or no?
Spatial (cues: over/under)	Blue is over green; green is over yellow	Speaker-relata: What is over? Blue or green? Speaker-relation: Is blue over or under yellow? Speaker-yes-no: Blue is over green, yes or no?	Speaker-relata: What is over? Green or blue? Speaker-relation: Is green over or under yellow? Speaker-yes-no: Green is over blue, yes or no?	Speaker-relata: What is over blue or yellow? Speaker-relation: Is blue over or under yellow? Speaker-yes-no: Blue is over yellow, yes or no?

Note. ME. Mutual-entailed relations. *CE.* Combinatorial-entailed relations. The table is only intended as a sample. A full list of the trials and assessment materials is available from the authors upon reasonable request.

response topographies during the relational assessment, we used a Friedman Test to evaluate if participants' scores across the response topographies were different. The Friedman test is the nonparametric version of the repeated measures ANOVA because the assumption of normality was not met.

3. Results

3.1. Descriptive summary of data obtained from the WCST and the relational assessment

During the WCST, on average, participants took 88.2 trials (SD = 17.9) to complete the test, emitted 71.5 (SD = 9.8) correct responses, and committed 7.8 (SD = 4.5) perseverative errors. During the relational assessment, on average, participants took 29.84 min (SD = 5.24) to complete the assessment and emitted 169.7 correct responses (SD = 20.21) out of a maximum possible score of 198. Across the entire sample, the overall rate of correct responses was 5.99/min (SD = 1.56) and the rate of correct responses per min by frame was 5.46/min coordination responses (SD = 1.73), 5.51/min distinction responses (SD = 1.78), 6.52/min comparison responses (SD = 2.11), 5.58/min temporality responses (SD = 2.16), 5.78/min hierarchy responses (SD = 1.93), 7.31/min spatial responses (SD = 1.99), and 5.79/min deictic responses (SD = 1.19). In terms of response topographies, out of a maximum score of 50 per topography, participants demonstrated an average of 42.79 correct speaker-relation responses (SD = 6.08) across all frames, 42.71 correct speaker-relata responses (SD = 5.87), and 41.76 correct speaker-yes-no responses (SD = 6.56). See Table 3 for the total duration to complete each frame and correct response rate, and see Appendix A for comparisons between neurotypical and neurodivergent participants. Given the small sample size, comparisons between the self-reported neurotypical and neurodivergent participants were not the primary aim of the study. Therefore, we included the group-wise descriptive data as part of the supplementary information (see more on this in the Discussion).

3.2. Differences between response topographies

To analyze differences between response topographies (i.e., speaker-relation responses, speaker-relata-responses, and speaker-yes-no-responses), we conducted a Friedman test. We found no statistically significant difference between the three response topographies $\chi2(2)$ = 5.41, p = .067.

3.3. Correlations between measures from the WCST and the relational assessment

We conducted a Spearman's non-parametric correlation analysis between all the WCST measures and the scores on the relational assessment. Table 4 shows the correlation coefficients for all comparisons. The analyses revealed a moderate negative correlation between the total trials taken to complete the WCST and rate of correct responses on the relational assessment $\rho = -.45$ [-.67, −.16], $p < .01$, $N = 42$. We did not find a statistically significant correlation between the total correct responses on the WCST and rate of correct responses on the relational assessment $\rho = -.32$ [-.56, .01], $p = .052$, $N = 42$.

Table 3
Participant scores on the arbitrary relational assessment and WCST.

Measure	Mean correct responses (SD)	Mean Total Duration min (SD)	Mean rate of correct responses/min (SD)
Frames			
Coordination	13.07 (2.18)	2.54 (.52)	5.46 (1.73)
Distinction	13.19 (2.4)	2.54 (.52)	5.5 (1.78)
Comparison	26.1 (4.38)	4.26 (1.07)	6.52 (2.11)
Temporality	23.52 (5.45)	4.53 (1.31)	5.58 (2.16)
Hierarchy	24.60 (4.43)	4.49 (1.41)	5.78 (1.93)
Spatial	26.79 (3.38)	3.86 (.87)	7.31 (1.99)
Deictic	42.45 (4.24)	7.53 (1.17)	5.79 (1.19)
WCST			
Total Trials	88.17 (17.88)		
Total Correct	71.48 (9.8)		
Perseverative Errors	7.79 (4.5)		
Perseverative Responses	8.36 (5.38)		

Note. Maximum score on coordination and distinction frames was 15, the deictic frame was 48, and 30 for all other frames.

Interestingly, we found statistically significant correlations between the perseverative errors during the WCST and the overall and frame-by-frame rate of correct responses on the relational assessment. Results revealed a strong negative correlation between perseverative errors in the WCST and rate of correct responses on the relational assessment, $\rho = -.65$ [-.80, −.42], $p < .01$, $N = 42$ and moderate to strong correlations between perseverative errors on the WCST and individual relational frames; coordination rate $\rho = -.57$ [−.75, −.31], $p < .01$; distinction rate $\rho = -.50$ [-.70, −.23], $p < .01$; comparison rate $\rho = -.64$ [-.80, −.42], $p < .01$; temporality rate $\rho = -.57$ [-.75, −.32], $p < .01$; hierarchy rate $\rho = -.59$ [-.76, −.34], $p < .01$; spatial rate $\rho = -.54$ [-.73, −.27], $p < .01$; and deictic rate $\rho = -.63$ [-.79, −.4], $p < .01$. Similarly, results revealed a strong negative correlation between perseverative responses in the WCST and rate of correct responses on the relational assessment, $\rho = -.64$ [-.8, −.41], p < .01, N = 42 and moderate correlations between perseverative errors on the WCST and individual relational frames; coordination rate $\rho = -.57$ [−.75, −.32], $p < .01$; distinction rate $\rho = -.52$ [-.72, −.25], $p < .01$; comparison rate $\rho = -.62$ [-.78, −.38], $p < .01$; temporality rate $\rho = -.55$ [-.73, −.28], $p < .01$; hierarchy rate $\rho = -.59$ [-.76, −.34], $p < .01$; spatial rate $\rho = -.53$ [-.72, −.26], $p < .01$; and deictic rate $\rho = -.62$ [-.78, −.36], $p < .01$.

4. Discussion

Our study was a preliminary attempt to evaluate whether there was any association between scores on the WCST and an experimenter-defined relational assessment in verbal adults. The resulting data showed a moderate negative correlation between the number of perseverative errors and perseverative responses during the WCST and the overall rate of correct responses during the relational assessment, and also moderate correlations between perseverative errors with individual relational frames. That is, participants with higher scores on the relational assessment made fewer perseverative errors and perseverative responses on the WCST. We also found a small to moderate correlation between the total trials taken to complete the card sort test and overall rate of correct responses on the relational assessment. Finally, we did not find a correlation between the number of correct responses during the WCST and the overall rate of correct responses on the relational assessment. We found no differences between the response topographies that we assessed during the relational assessment. To the best of our knowledge, this is the first study to investigate associations between an EF measure and relational framing in verbal adults. This is especially relevant given that EF has been shown to predict success and well-being both in children and adults (Morgan et al., 2018; Toh et al., 2020).

Our correlational analyses showed overall that perseverative errors committed during the WCST were correlated with all individual frames and with overall relational assessment scores. Relatively, although the difference was small, the comparison and deictic frames were more strongly correlated with perseverative errors than the other frames in our study. Tyrberg et al. (2021) assessed participants' verbal self-talk during the WCST and found that cues specifying deictic and temporal frames were more prevalent during rule-changes. In our study, although both temporality and deictic scores were significantly correlated with perseverative errors, we found that deictic scores revealed a relatively higher correlation coefficient ($\rho = .63$ for deictic vs. .57 for temporality). We also found a relatively stronger correlation coefficient for the comparison frame ($\rho = .64$). It seems reasonable to expect that conditional/causal relations may facilitate correct responding after an unannounced rule change. For instance, an individual may frame the rule-changes relationally by saying, "Matching by color is incorrect, so I should try matching by shape". Such causal framing specifies a functional relation between the feedback from the computer and how this could be changed. An important next step for future research may be to include tests for causal/conditional frames during the relational assessment.

As discussed earlier, while there are multiple measures of EF, each assessing specific components, the WCST tests one component called set-shifting (i.e., behavior change during unannounced rule changes). From an RFT-perspective, one could argue that relational framing was most relevant during rule changes to facilitate identification of the new rule (Hayes et al., 1996). For example, successfully identifying a new rule may involve observing the negative feedback and answering to oneself about the previous rule and making conjectures about other rules to try. One's relational framing abilities (e.g., "before", "I/Now/Then", "different") may be controlling the behavioral responses emitted during

Table 4
Correlation matrix of WCST and arbitrary relational assessment.

		1	2	3	4	5	6	7	8	9	10	11	12
1.	WCST Total Trials												
2.	WCST Correct	.91**											
3.	WCST Pers. Error	.75**	.63**										
4.	WCST Pers. response	.76**	.63**	.99**									
5.	AARR Score	−.42**	−.35*	−.54**	−.54**								
6.	AARR Rate	−.45**	−.30	−.65**	−.64**	.78**							
7.	Coordination	−.44**	−.25	−.57**	−.57**	.63**	.74**						
8.	Distinction	−.35*	−.17	−.50**	−.52**	.56**	.71**	.96**					
9.	Comparison	−.40**	−.26	−.64**	−.62**	.76**	.91**	.71**	.65**				
10.	Temporality	−.45**	−.36*	−.57**	−.55**	.71**	.83**	.50**	.49**	.74**			
11.	Hierarchy	−.44**	−.34*	−.59**	−.59**	.75**	.91**	.59**	.56**	.82**	.75**		
12.	Spatial	−.34*	−.20	−.54**	−.53**	.61**	.88**	.53**	.53**	.81**	.75**	.79**	
13.	Deictic	−.37*	−.21	−.63**	−.62**	.45**	.64**	.32*	.25	.59**	.58**	.52**	.54**

Note. *p < .05; **p < .01. For all individual frames, we used the rate of correct responses/min for the analyses. WCST Wisconsin Card Sort Test. AARR Arbitrarily applicable relational responding. Pers. Perseverative.

the WCST, specifically when the rules changed. Stated in other words, performing well on the test may involve responding to ambiguity, i.e., the unannounced rule changes with over-arching verbal strategies (Hayes et al., 2001). Presumably the correlations we found could indicate that correct derived relational responding as demonstrated during the relational assessment generalizes to ambiguous problem-solving situations such as the WCST. Nevertheless, since we only found a correlation, it is plausible that high levels of set-shifting behavior facilitate verbal activity in the form of derived relational responding, or some other variable facilitates both set-shifting and relational responding.

We found no correlation between the total correct responses on WCST and the overall scores on the relational assessment. Given the wider consistent pattern of correlations, this must be viewed tentatively. Nevertheless, the total number of correct responses on the WCST is a measure obtained by adding the total number of correct matching responses across the entire test. Specifically, the test is designed such that an individual must emit ten consecutive correct matching responses after which the system changes the matching rule (i.e., after ten successful trials of matching by color, the system changes the rule to matching by shape) and the individual must now emit ten consecutive correct responses based on the new rule and so on. Therefore, an overwhelming majority of these responses involve matching after the rule has been identified. Once a rule has been identified, correct responses simply involve following the rule to complete a nonarbitrary match-to-sample task. Arguably, these matching responses need not involve complex verbal behavior for successful completion, and indeed even some nonhuman animals have shown success in matching compound stimuli based on physical features such as shape or brightness (e.g., Hopkins & Washburn, 2002). One's relational framing may facilitate rule identification but does not seem to be necessary for successively emitting correct matching responses after the rule has been identified.

The findings we present herein need further replication and extension with children and adults. In order to demonstrate a functional relation, future studies could aim to strengthen relational skills (e.g., Cassidy et al., 2011; Stricker et al., 2024) and assess pre- and post-intervention scores on EF measures such as the WCST. Such work could follow a similar research agenda to the SMART training studies that use intelligence as a dependent measure (e.g., Colbert et al., 2019; Roche et al., 2023). In terms of building bridges with the cognitive psychology literature, our findings are broadly consistent with that of Starr et al. (2023) who found that in a sample of 942 children, scores on a nonarbitrary analogy test (called relational thinking by the authors) were strongly correlated with three composite EF scores and uniquely predicted variance in mathematics scores. These authors recommended that relational responding may be considered a type of EF in itself. Similarly, Todd et al. (2019) measured relational thinking in 125 older adults using a series of tasks, one of which was a comparison test involving "more than" and "less than" relations between three-, four- and five-member classes with arbitrary stimuli, and they found that accuracy during the relational tasks was correlated with other executive functioning scores. While none of these studies examined relational activity in sufficient detail as afforded by RFT (see Colbert et al., 2017; Cummins, 2023, for examples), they present an important avenue for collaboration. Studies that attempt to measure derived relational responding as defined by RFT along with traditional EF tests could advance both cognitive and behavioral psychology by (a) identifying relational responding that facilitates performance during specific EF tasks thereby increasing the scope of RFT (see O'Toole et al., 2009, for an example of such work connecting relational activity to various intelligence tests), and (b) offer a behavioral basis for executive functioning ultimately leading to targeted behavioral interventions for specific types of cognitive challenges in children and adults (see Cassidy et al., 2010, for a behavioral interpretation of human intelligence, and Cassidy et al., 2016 for a training study to improve IQ scores).

Readers should note that we had a small sample of neurodivergent participants, we did not verify their diagnoses, and we simply used their self-reports. There was also wide variation in the self-reported diagnoses from characteristics of autism and ADHD to a learning disability with lower individual sample sizes for each of these conditions within the broad neurodivergent sample (but see Hervey et al., 2004, for a meta-analysis showing that the WCST was not effective at discriminating between groups with and without ADHD characteristics). Given these limitations, we chose to include the group-level data as part of supplementary information. Future research should strive to include a larger and more diverse sample of neurodivergent individuals, include their specific diagnostic information, and analyze differences (if any) between these groups. This seems particularly important considering that some neurodivergent adults do not readily demonstrate fluent derived relational responding (e.g., Zagrabska-Swiatkowska et al., 2020) and have been shown to emit more perseverative responses than neurotypical participants during tests of set-shifting (e.g., Tsuchiya et al., 2005; Westwood et al., 2016; Yasuda et al., 2014).

Although there are other studies that have used the Relational Abilities Index (RAI) as a comprehensive measure of relational activity (e.g., Colbert et al., 2019; Cummins, 2023), we adapted the relational assessment used by Kirsten & Stewart (2022) and included distinct response topographies during the relational assessment. We chose to use the latter because we aimed to test if there were any differences between speaker-topographies measured during the assessment (the RAI typically only assesses selection responses). But we found no significant differences between these topographies. As noted in the introduction, we expected to see participants emit fewer correct yes/no responses compared to the other response topographies, but we did not find this pattern.

Our analyses were limited in that (a) we did not experimentally verify if self-generated rules controlled the participants' matching responses during the WCST (e.g., using the silent dog method; Hayes et al., 1998, or pragmatic verbal analysis; Hayes et al., 2001), (b) we did not measure conditional/causal relations during the relational assessment, (c) we did not use any other measures of EF besides the WCST, and (d) we did not include a clinically verified sample of individuals who experienced challenges with set-shifting. Given the everyday relevance of responding flexibly to a changing set of societal contingencies, this line of work could have key practical applications ultimately resulting in the development of robust training programs that impact set-shifting in the natural environment in children and adults who experience challenges in this area. We hope that the current pilot study serves to inspire more research in this area and extends the scope of RFT by explaining phenomena such as EF, which is said to facilitate reasoning, self-regulation, and problem-solving, all key abilities for individuals' success and well-being.

CRediT authorship contribution statement

Xiaoyuan Liu: Writing – review & editing, Writing – original draft, Project administration, Methodology, Data curation, Conceptualization. **Maithri Sivaraman:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Data curation, Conceptualization. **Elle Kirsten:** Writing – review & editing, Conceptualization.

Ethics

The study was conducted ethically with approval obtained from the Institutional Review Board of the first author's institution. All participants provided written informed consent prior to participation in the study.

Data availability statement

Anonymized data is available upon reasonable request from the first author.

Conflict of interest

All the authors are practicing behavior analysts and stand to benefit professionally from scientific reports attesting to the use of applied RFT. Additionally, Dr. Elle Kirsten is the CEO of RYIQ and the founder and director of Compassionate Behavior Analysis, PLLC (CBA). The findings in this study provide evidence for RFT-based language interventions in general which may indirectly benefit RYIQ and CBA. We have no direct conflicts of interest to disclose.

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Appendix A. Participant Scores on the Arbitrary Relational Assessment and WCST

Measure	Neurotypical participants			Neurodivergent participants		
	Mean correct responses (SD)	Mean Total Duration min (SD)	Mean rate of correct responses/min (SD)	Mean correct responses (SD)	Mean Total Duration min (SD)	Mean rate of correct responses/min (SD)
Frames						
Coordination	13.58 (1.62)	2.49 (.51)	5.76 (1.67)	11.2 (3.00)	2.7 (.53)	4.39 (1.47)
Distinction	13.55 (2.28)	2.49 (.51)	5.75 (1.8)	11.9 (2.52)	2.7 (.53)	4.63 (1.47)
Comparison	26.39 (4.55)	4.08 (1.00)	6.87 (2.22)	25 (3.74)	4.91 (1.1)	5.21 (.82)
Temporality	24.22 (5.27)	4.34 (.99)	5.97 (2.22)	20.56 (5.37)	5.2 (2.05)	4.16 (1.11)
Hierarchy	25.45 (4.06)	4.44 (1.10)	6.13 (1.92)	21.44 (4.56)	5.18 (2.2)	4.51 (1.44)
Spatial	27.2 (3.09)	3.79 (.89)	7.59 (2.07)	25.22 (4.12)	4.11 (.79)	6.26 (1.22)
Deictic	46 (7.62)	7.5 (1.29)	5.97 (1.22)	41.11 (4.73)	7.63 (.55)	5.12 (.75)
WCST						
Total Trials	85.91 (17.23)			96.44 (18.78)		
Total Correct	70.45 (9.3)			75.22 (11.22)		
Perseverative Errors	7.3 (4.5)			9.7 (4.3)		
Perseverative Responses	7.8 (5.4)			10.4 (5)		

Note. Maximum score on coordination and distinction frames was 15, the deictic frame was 48, and 30 for all other frames.

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